

NanoGT Match Report: Leber hereditary optic neuropathy

Disease: Leber hereditary optic neuropathy (ORPHA:104)

Primary gene: MT-ND4

Gene CDS: 1377 bp

Inheritance: Mitochondrial inheritance

Target tissues scored: retina, CNS

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.; Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints; Vector does not naturally cover all annotated disease tissues: cns, retina.

Disease Mechanism Evidence

Molecular mechanism: mitochondrial loss of function

Mechanistic detail: Mitochondrial complex I dysfunction from MT-ND4 variant

Gene-addition compatibility: uncertain

Preferred modality class: specialized allotopic expression or mitochondrial strategy

Evidence level/status: direct / needs_manual_review

Evidence summary: Mitochondrial DNA disease is not ordinary nuclear gene addition; allotopic AAV approaches are specialized and disease-specific

Evidence source: OMIM MT-ND4 gene entry

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Endpoint readiness: retinal diseases often have measurable endpoints such as OCT, ERG, visual acuity, visual fields, or mobility testing, but genotype-specific progression still needs confirmation.
 - Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
 - Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	CPCB-RPE1	AAV8	6.4/10	Medium	phase2/3
2	Skysona	LV	6.3/10	Medium	approved
3	Luxturna	AAV2	6.1/10	Medium	approved
4	Libmeldy	LV	5.9/10	Medium	approved
5	Hemgenix	AAV5	5.6/10	Medium	approved

Match #1: CPCB-RPE1

Precedent disease: Achromatopsia

Vector: AAV8

Tissue target: retina/photoreceptor

Composite score: 6.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.70	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	0.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.43	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1377bp / cargo 4700bp (29% utilized)
- Precedent target match: retina
- Both membrane proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Pathway match: retinal_phototransduction
- Disease mechanism: mitochondrial loss of function — Mitochondrial complex I dysfunction from MT-ND4 variant
- Gene-addition modality compatibility: uncertain for ordinary gene addition
- Mechanism evidence: Mitochondrial DNA disease is not ordinary nuclear gene addition; allotopic AAV approaches are specialized and disease-specific
- Mechanism source: OMIM MT-ND4 gene entry (<https://omim.org/entry/516003>)
- Approval status: phase2/3
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: INCOMPATIBLE — MT-ND4 is a mitochondrial DNA-encoded gene. Standard nuclear AAV delivery cannot produce a functional protein at the mitochondrial target site. Treatment requires allotopic expression: cytoplasmic recoding of the gene using the standard genetic code plus an artificial mitochondrial targeting sequence (MTS). This strategy is disease-specific and fundamentally different from the nuclear gene-addition programs in this catalog. All precedent scores for this disease should be treated as cross-paradigm comparisons only, not direct development templates. (Real-world precedent: GS010/Lumevoq for MT-ND4/LHON; Phase 3 completed, EMA MAA withdrawn April 2023.)

Manual Review Flags

- MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns, retina
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology

- Mechanism evidence does not cleanly support simple gene addition; consider RNA, editing, silencing, mitochondrial, or other non-catalog modalities
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	0.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.33	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1377bp / cargo 8000bp (17% utilized)
- Precedent target match: cns
- Both membrane proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (retinal_phototransduction vs peroxisomal)

- Disease mechanism: mitochondrial loss of function — Mitochondrial complex I dysfunction from MT-ND4 variant
- Gene-addition modality compatibility: uncertain for ordinary gene addition
- Mechanism evidence: Mitochondrial DNA disease is not ordinary nuclear gene addition; allotopic AAV approaches are specialized and disease-specific
- Mechanism source: OMIM MT-ND4 gene entry (<https://omim.org/entry/516003>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: INCOMPATIBLE — MT-ND4 is a mitochondrial DNA-encoded gene. Standard nuclear AAV delivery cannot produce a functional protein at the mitochondrial target site. Treatment requires allotopic expression: cytoplasmic recoding of the gene using the standard genetic code plus an artificial mitochondrial targeting sequence (MTS). This strategy is disease-specific and fundamentally different from the nuclear gene-addition programs in this catalog. All precedent scores for this disease should be treated as cross-paradigm comparisons only, not direct development templates. (Real-world precedent: GS010/Lumevoq for MT-ND4/LHON; Phase 3 completed, EMA MAA withdrawn April 2023.)

Manual Review Flags

- MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns, retina
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence does not cleanly support simple gene addition; consider RNA, editing, silencing, mitochondrial, or other non-catalog modalities

Match #3: Luxturna

Precedent disease: Leber congenital amaurosis type 2

Vector: AAV2

Tissue target: retina/RPE

Composite score: 6.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	0.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	0.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.10	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1377bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: retina
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Pathway match: retinal_visual_cycle
- Disease mechanism: mitochondrial loss of function — Mitochondrial complex I dysfunction from MT-ND4 variant
- Gene-addition modality compatibility: uncertain for ordinary gene addition
- Mechanism evidence: Mitochondrial DNA disease is not ordinary nuclear gene addition; allotopic AAV approaches are specialized and disease-specific
- Mechanism source: OMIM MT-ND4 gene entry (<https://omim.org/entry/516003>)
- Approval status: approved
- Vector immunogenicity (AAV2): very high (~55%) — majority of patients may be ineligible; major trial design challenge

- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: INCOMPATIBLE — MT-ND4 is a mitochondrial DNA-encoded gene. Standard nuclear AAV delivery cannot produce a functional protein at the mitochondrial target site. Treatment requires allotopic expression: cytoplasmic recoding of the gene using the standard genetic code plus an artificial mitochondrial targeting sequence (MTS). This strategy is disease-specific and fundamentally different from the nuclear gene-addition programs in this catalog. All precedent scores for this disease should be treated as cross-paradigm comparisons only, not direct development templates. (Real-world precedent: GS010/Lumevoq for MT-ND4/LHON; Phase 3 completed, EMA MAA withdrawn April 2023.)

Manual Review Flags

- MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence does not cleanly support simple gene addition; consider RNA, editing, silencing, mitochondrial, or other non-catalog modalities
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 5.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity

Dimension	Score	Max	What it measures
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	0.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.86	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1377bp / cargo 8000bp (17% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (retinal_phototransduction vs leukodystrophy)
- Disease mechanism: mitochondrial loss of function — Mitochondrial complex I dysfunction from MT-ND4 variant
- Gene-addition modality compatibility: uncertain for ordinary gene addition
- Mechanism evidence: Mitochondrial DNA disease is not ordinary nuclear gene addition; allotopic AAV approaches are specialized and disease-specific
- Mechanism source: OMIM MT-ND4 gene entry (<https://omim.org/entry/516003>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- β 2; minimal

T-cell clearance risk

- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: INCOMPATIBLE — MT-ND4 is a mitochondrial DNA-encoded gene. Standard nuclear AAV delivery cannot produce a functional protein at the mitochondrial target site. Treatment requires allotopic expression: cytoplasmic recoding of the gene using the standard genetic code plus an artificial mitochondrial targeting sequence (MTS). This strategy is disease-specific and fundamentally different from the nuclear gene-addition programs in this catalog. All precedent scores for this disease should be treated as cross-paradigm comparisons only, not direct development templates. (Real-world precedent: GS010/Lumevoq for MT-ND4/LHON; Phase 3 completed, EMA MAA withdrawn April 2023.)

Manual Review Flags

- MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: cns, retina
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence does not cleanly support simple gene addition; consider RNA, editing, silencing, mitochondrial, or other non-catalog modalities

Match #5: Hemgenix

Precedent disease: Hemophilia B

Vector: AAV5

Tissue target: liver

Composite score: 5.6 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway

Dimension	Score	Max	What it measures
Modality compatibility	1.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	0.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.62	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1377bp / cargo 4700bp (29% utilized)
- Vector tropism overlaps cns, but precedent target is liver
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (retinal phototransduction vs coagulation)
- Disease mechanism: mitochondrial loss of function — Mitochondrial complex I dysfunction from MT-ND4 variant
- Gene-addition modality compatibility: uncertain for ordinary gene addition
- Mechanism evidence: Mitochondrial DNA disease is not ordinary nuclear gene addition; allotropic AAV approaches are specialized and disease-specific
- Mechanism source: OMIM MT-ND4 gene entry (<https://omim.org/entry/516003>)
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: INCOMPATIBLE — MT-ND4 is a mitochondrial DNA-encoded gene.

Standard nuclear AAV delivery cannot produce a functional protein at the mitochondrial target site. Treatment requires allotopic expression: cytoplasmic recoding of the gene using the standard genetic code plus an artificial mitochondrial targeting sequence (MTS). This strategy is disease-specific and fundamentally different from the nuclear gene-addition programs in this catalog. All precedent scores for this disease should be treated as cross-paradigm comparisons only, not direct development templates. (Real-world precedent: GS010/Lumevoq for MT-ND4/LHON; Phase 3 completed, EMA MAA withdrawn April 2023.)

Manual Review Flags

- MITOCHONDRIAL DNA GENE (MT-ND4): this gene is encoded in the mitochondrial genome, translated by mitochondrial ribosomes using a non-standard genetic code. Standard nuclear AAV gene addition CANNOT produce this protein. Treatment requires allotopic expression: the gene must be recoded for cytoplasmic translation and given an artificial MTS — a strategy fundamentally different from every program in this catalog. All precedent scores are cross-paradigm comparisons. See: GS010/Lumevoq (Gensight Biologics) as a real-world allotopic precedent.
- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: retina
- Only partial tissue match; verify target-cell transduction and delivery route manually
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence does not cleanly support simple gene addition; consider RNA, editing, silencing, mitochondrial, or other non-catalog modalities
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone